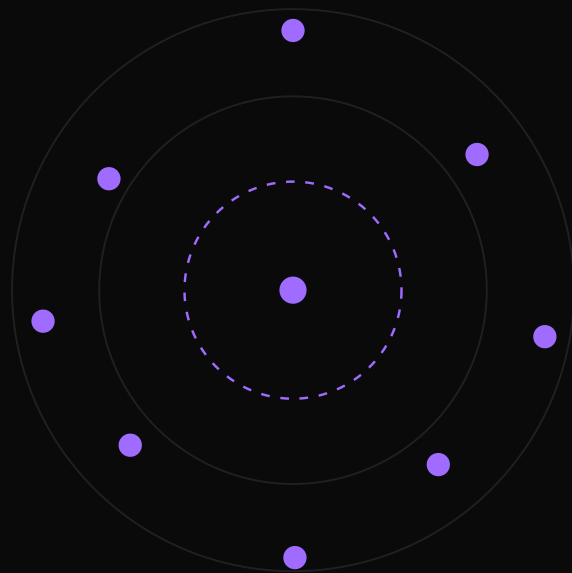


Which cells can sleep tonight?

A combinatorial optimisation problem from True's Bangkok network, mapped onto the geometry of a neutral-atom quantum processor — and run live on 256 atoms in Boston.

DELIVERED BY

QTRiC × qBraid



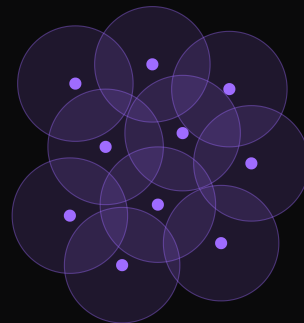
WHAT WE ARE SOLVING

Ten towers. One question, asked every night.

Tonight, at around 3 a.m., almost every cell site in this district will be awake and drawing close to full power — serving almost nobody. Including the ones on the roof of this building.

So every night a controller faces one decision per cell: sleep, or stay awake. The decisions are not independent — a cell can only sleep if a neighbour is still covering the ground beneath it.

- **Get it wrong for one cell** and a street loses signal at 3 a.m.
- **Get it too cautious** and the network burns power serving nobody.
- **Get it right** and the customer notices nothing at all.



Where the circles overlap, one tower can cover for another. Where they don't, nobody can.

For ten towers there are 1,024 ways to answer. Only 63 of them leave every square metre covered. The best one switches off four.

And Sukhumvit is not ten towers. One district is a number of combinations with more digits than there are seconds since the universe began — and the right answer changes every night as traffic moves.

FORMULATION

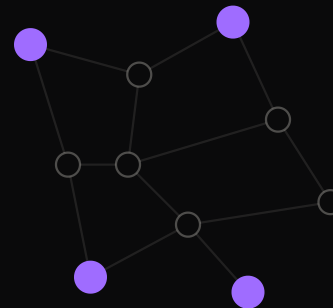
This question already has a name.

Replace each overlap with a line between two towers. The picture becomes a graph, and the question becomes a classical one: find the largest set of towers with no line between any two of them. Mathematicians call it **Maximum Independent Set**, and it has been studied for fifty years.

It is also known to be NP-hard. Solving it exactly requires global information about the whole cluster at once — which is precisely why operators fall back on per-cell rules.

Why this is the hard part

A rule that looks at one cell at a time has no view of the pair. It cannot know that switching off two neighbours simultaneously opens a hole between them. Seeing the pair requires holding the whole cluster in view at once — and that is the expensive part.



● asleep ○ awake

THE MAPPING

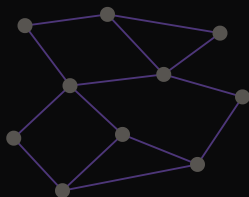
That shape is also the shape of the machine.

On a neutral-atom processor, two atoms within the Rydberg blockade radius physically cannot both be excited. That is not a penalty term we added. It is the coverage constraint, enforced by rubidium.

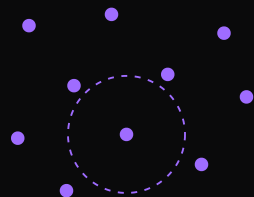
SUKHUMVIT SITES



COVERAGE GRAPH



ATOM REGISTER

 $1 \mu\text{m} = 60 \text{ m}$

the scale factor

 $8.4 \mu\text{m}$ blockade radius $\approx 500 \text{ m}$ of coverage overlap $4 \mu\text{m}$ minimum atom spacing $\approx 238 \text{ m}$ 4.5 km

of Sukhumvit, in an area narrower than a hair

$R_b = (C_6/\Omega)^{1/6}$ with $C_6 = 5,420,503 \mu\text{m}^6 \cdot \text{rad}/\mu\text{s}$ and $\Omega = 15.6 \text{ rad}/\mu\text{s}$ · Sarkar et al., arXiv:2511.09633 · We did not tune the blockade radius to fit; this is where rubidium sits

THE PROBLEM

At 3 a.m. the network is awake.
Almost nobody is.

Mobile traffic in Bangkok collapses overnight, but the radio access network keeps running at close to full readiness. This is where telecom energy actually goes — and it is the largest controllable cost line an operator has.

Every operator knows this. Almost none of them act on it aggressively. The next two slides are about why.

~90%

of an operator's energy is
consumed by the network

>80%

of that network energy sits in the
RAN

~23%

of total operator cost is energy

4 kW

typical draw per macro site, around
the clock

WHY NOW

Sleeping cells is already in the standard.
Sleeping them *together* is not.

/ 01

It is standardised

3GPP Release 18 made Network Energy Savings normative — cell DTX/DRX in TS 38.300 §15.4. This is spec, not research.

/ 02

It works

Vodafone UK with Ericsson, London 2025: AI features cut daily 5G radio-unit power by up to 33%, and deep sleep saves up to 70% in low-traffic hours.

/ 03

True is already ahead

Up to 20% energy reduction across network sites using AI and ML, 17,000+ base stations integrated under One Network, plus the Genie platform.

Every one of these decides cell by cell. A per-cell rule cannot see that switching off two neighbours at the same time opens a hole between them. That joint decision is the part nobody solves — and it is the part that is NP-hard.

THE REAL BLOCKER

The constraint is not compute. It is proof.

No operator switches off radios in a capital city on a heuristic's say-so. The downside is asymmetric: a coverage hole is a dropped emergency call, a regulator's letter, and a headline.

True has already been publicly reprimanded by the NBTC over site changes that customers blamed for poor signal. After that, the tolerance for algorithmic risk in the Bangkok RAN is close to zero — reasonably so.

So the question is not "can we compute a better sleep set." It is **can we hand the network team a guarantee they can sign off on.**

OUR SAFETY RULE

A cell may sleep only if a neighbour is awake to cover its footprint.

Enforced conservatively, this becomes: no two mutually-covering cells may sleep simultaneously. A solver that respects it returns not just an answer but a **certificate** — provably no hole.

That certificate is what lets True turn the dial from cautious to optimal without a regulatory incident. It is the entire value proposition.

THE EXPERIMENT

Sukhumvit, on 150 atoms.

/ 01

Data

OpenCellID, Watthana district.
N cells clustered into N sites,
merged to 150 atoms at the 238
m floor.

/ 02

Device

QuEra Aquila, 256 neutral rubidium
atoms, Boston. Submitted through
qBraid Lab.

/ 03

Method

Quasi-adiabatic sweep: detuning
from negative to positive over
4 μ s, the coherence budget of
the machine.

/ 04

Statistics

3 tasks $\times$ 1,000 shots, the per-
task ceiling. Best-of-shots
reported with the full distribution.

And the 20-atom instance you are running right now on your own machine is the same pipeline, same code, same physics — just small enough to finish while I am talking.

RESULT

41 cells asleep. No coverage hole.

41 / 150

cells the machine says can sleep

0.9xapproximation ratio vs exact
classical**000 MWh**

annualised, this district

฿ 0.0 M

annualised, this district

PLACEHOLDER

Sukhumvit map with the sleeping cells
highlighted, generated from actual
Aquila measurement outcomes. Insert
after the pre-run.

Coverage verified. Every sleeping cell in this set has all of its
overlapping neighbours awake. That is checked in the notebook and
printed, not asserted on a slide.

HONEST SCOREBOARD

Classical won today. Here is by how much.

We are not claiming quantum advantage, and we would not survive this room if we did. A generic branch-and-bound solver finds the exact optimum on instances of this size in under a second.

SITES	EXACT SOLVE, COMMODITY HARDWARE
60	0.009 s
120	0.103 s
180	3.31 s
Aquila, 150 atoms	000 s

What the literature says

Andrist et al. (2023) solved quasi-planar unit-disk instances to optimality up to thousands of nodes within minutes on commodity hardware, and found relaxed simulated annealing competitive with the quantum algorithms.

Cazals et al. (2026), from a neutral-atom hardware company, report that on small instances quantum is slower than classical for exact solutions.

So why run it at all? Because the encoding is different in kind: no penalty weights, no infeasible answers to repair. And because the only way to know when this crosses over is to measure the gap every year. That is what an MOU is for.

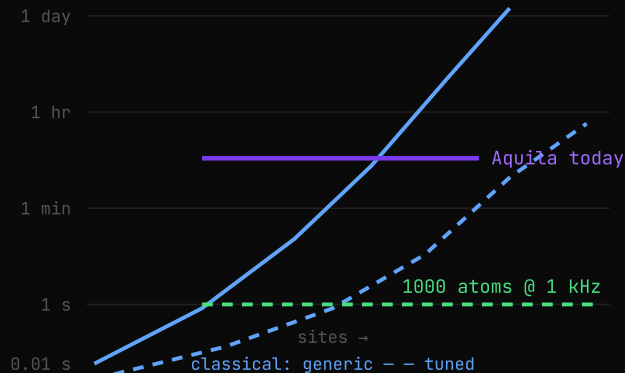
THE GAP

The missing number is not qubits.
It is shot rate.

Classical solve time grows exponentially with problem size. Analog quantum runtime is **flat** in problem size — the same $4\ \mu\text{s}$ sweep whether you place 50 atoms or 250. So this is not two curves racing. It is an exponential crossing a horizontal line.

Two things set where they meet: how low the flat line sits, which is shots divided by repetition rate, and where it stops, which is atom count.

Atom count is nearly solved — coherent operation of 3,000 physical qubits with automated reloading has been demonstrated. Going from a few hertz to a kilohertz is two to three orders of magnitude on a metric that appears on almost no public roadmap.



Our estimate: 2029–2032, wide error bars, and conditional on real Bangkok instances being hard. The classical baseline moves too.

THE VALUE

\$30–75 million a year. Available now.

Built entirely on True’s own published constants, so only the physical assumptions are arguable, not the conversion.

ANCHOR	VALUE
True’s AI/ML energy saving, 2023	42,000 MWh
Booked as	\$197 million
Implied tariff	\$4.69 / kWh
Implied grid intensity	0.45 tCO ₂ e / MWh

EXTRA CELLS SLEPT VS PER-CELL RULES	THAILAND
+10 points	\$30M / yr
+15 points	\$44M / yr
+25 points	\$74M / yr

+23%

on the \$197M True already banks from AI energy work

~4,250

tonnes CO₂e per year at the midpoint

And the uncomfortable part. Every baht of this is reachable today with a classical solver. Quantum’s contribution to this number in 2026 is zero. We are showing you the money to prove the problem is worth solving — and the benchmark to show you when the machine catches up.

ASSUMPTIONS

What we are assuming, and who can falsify it.

- **4 kW per macro site.** Our largest single lever on the money model. True's network team can replace this in an afternoon.
- **Genie decides per cell, not jointly.** Our central hypothesis. If True already solves the joint subset problem, the incremental value collapses.
- **Coverage as uniform disks.** Real radii vary by band, tilt and cell class. Standard in the literature; still an approximation.
- **Independent set, not dominating set.** A conservative relaxation. Always feasible, but it may leave savings unclaimed.
- **Crowdsourced positions.** OpenCellID estimated centroids, not surveyed towers. Sites within 238 m are merged to fit the atom spacing floor.
- **No quantum advantage claimed.** On these sizes classical wins. We are measuring a gap, not declaring a victory.

Every one of these is printed in the notebook you can run yourself. We would rather you find the weak assumption today than in a pilot review six months from now.

PATH FORWARD

Apply, don't build.

"Rather than building quantum computers, we will focus on applying quantum technologies to create value in industries where Thailand already has strengths." — the National Quantum Economy Strategy, approved 24 June 2026.

/ 01 PILLAR

Industrial application

A 90-day pilot on real True site data. Classical solver in production, quantum benchmark alongside. Measured in baht and tonnes.

/ 02 PILLAR

The annual benchmark

True, QTRiC and qBraid co-author the crossover study — instance hardness, shot rate, approximation ratio. Published, repeated yearly.

/ 03 PILLAR

Workforce

This notebook opens in a browser. Any student at any Thai university can run tonight's experiment tomorrow morning, on the same hardware.

Thailand does not need to build a quantum computer to lead on this. It needs one industry, one honest benchmark, and the people who can read it. We are proposing to start with the network under this building.